

Social constraints (Discussion) ~ Avel Guénin-Carlut

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hello welcome everyone it is October 25th 2023 we're in active inference for the social sciences and today we're in a discussion section it is AAL and I and we'll start off by just saying hello and kind of reflecting on where we're at with the course then we have the questions form that is on the course website it's in the video description feel free to submit questions via this form and then today we're going to look at the questions that have been submitted and not answered yet and we'll just give some thoughts on them also look for more questions to arise in the live chat so with that what do you want to begin with AVL feel free to give any kind of overview or statement and then we'll jump to the specific questions that have been submitted before today and during today the only statement that I want to make is the last presentation I did so prediction cause constraints to exist would be our wealth uh sorry of that but I really I don't have anything to say that he's more specific right now all right well this should be pretty fun let's jump into some questions okay what are the challenges and potential Solutions in using active inference to model Collective behaviors with various environmental constraints [Music] um uh so the uh challenges first is that in any model you have to pick what you include and what you did not include so when you talk about boundary conditions when you talk about what's including the model you have to make hypothesis like if I tell you model I don't know country that is under sanctions like North Korea Russia okay how do you model that do you make a state Bas for every possible product like every Dimension accounts for a specific product or type of product and you have zero if you have access to sorry a zero if you don't have access to it and one is you access to it okay but what if we have new things that happen that that that is not very useful if you're interested is an innovation or something so we will have to make specific par that uh will traduce specific questions that you want to ask the model and using active INF front has no like does not change anything to that and where it is relevant is that um you can have uh endogenous evolution of constraints using Acer I think in

a more meaningful sense than you have with a pure let us say hean or like non representational uh framework or framework that is presentational but where the space representation is like hardcoded so you there is a space of models let us say of evolution of social norms where I think active inference would have a a nedge because active inference is built to represent normativity and intentionality and it is not the case to my knowledge of other like either dynamicist or cognitivist um Frameworks for modeling could you maybe restate and go into a little more detail was it non-representational or indirect specification of what of the um beliefs of the system so what's being directly specified what's directly specified and then what is being indirectly specified inactive inference yeah inactive inference you have at least in a sense uh correspondence between dynamical constraints like information over where system is and belief that the system holds that are akin to Norms or like inal States so beliefs or intentions are things that are going on inside and to my knowledge there is no framework that makes an explicit foral connection between the two You' have damis framework uh where you have stuff that happens and there is no presentation or intentional States and you have things like utility Theory where you have to specify uh preemptively what are the interntional states of the agents which are um not very relevant toes um like the enous evolution of social constraints within a collective system uh so um situation where active inference would be useful is to model I uh I will try to remember the term that kofman used kofman Fillin and Fillin in um so I will share the can can we both share the screen no we can't you can add it directly in in this um Coda page probably a little easier than all just go to the backend course Coda page and then follow the token yep and then go to the questions table and pop it out you don't have the questions that is on your screen just click on little um uh challenges and potential Solutions and then pop it out pop out what um just like next to the row see this expand row and then you'll be in the same place what is the row C there is no letter no no no no just look visually at the expand row look visually at the expand row just click the expand row on the question that we're addressing and you'll pop into this campus the question that we are addressing is not on the page at least on my terminal so are you looking at the public front end or at the course back end I'm looking at the back end okay and I do not have the question what are blah blah blah oh it is a sub question what other CH she is okay okay cool all right fun you were talking about Kaufman though what was it yes it's a paper that the cosigned was manator isin critic of economic uh like I say rationality but now they say it's Bing rationality so the um dictionary of cognitive biases approach to modeling human behavior and uh basically the point of the paper as I understand it is that you have an evolution of the goals and

Norms uh that shape a social ecology and this kind of open-ended uh evolution of the intentional state of the agent I think is can be addressed in a more meaningful way with Acer than with any other formal framework I of there are several other aspects I think so first to kind of restate the the challenges at then go to the potential Solutions and the advantages um like the advantages in principle and in practice you started with identifying like the major any kind of modeling challenges so ACM inherits from all of these modeling challenges or areas that that are difficult to step aside from and then yes depending on the resources of the group and whether it's 2023 or 2 you know future time when there's different tooling so that is just about the the challenge of active inference sometimes compared today with more off-the-shelf approaches which also may or may not work but in an acadm you'll be spending your time making something that may or may not work may or may not be orders of magnitude more just sophistication of a modeling setting than the team is ready to again that's a General modeling point but it's also very easy to get into with ACM and to jump the gap between something that is pedagogical and helps show something like intuitive or cool and making something for example that can be used in high reliability settings and that's a jump that hasn't been necessarily conveyed in the open source setting so it's difficult to see examples of whereas there's kind of like production type machine learning algorithms that can at least be no that's just a little general point so what are some advantages there's some really interesting concepts related to this like indirect specification so like let's just say that we were talking about the mountain car example or a social variable that seemed to be moving within a certain range then the direct encoding of that can strain would um be basically made outdated okay now the landscape changes and then all of a sudden the mountain car can go twice as far to a different direction so if that constraint had been explicitly encoded about cognitive um variables or about the exposition of the mountain car then the model goes stale but if what's encoded is somehow a relation with the agent in the environment so the Mountain car has 10 horsepower and it can go left or right as its affordances and everything that builds from there and then here's the laws of physics or mechanics such that here's the weight of the car and and then if the landscape changes or his model is changing then you get oh yeah now it can go further or shorter in this direction but that's like an outcome of the relationship between the agent in the environment explicitly represented as such versus just looking at the outcome and then putting constraints on the outcome yes uh to say we have uh historically the concept of explanation that we have inherited from the uh 19th 20th century is uh that we explain by having a Love full relationship between variables and to do that we have to press specify what

are the relevant variables and this is not the thing that exist with open-ended evolutionary Dynamics this is all in the um Galu ctive agency something something paper so um and so uh when we ask questions to a model we don't want to have the answer yeah it will evolve ously we don't really know this is not what is expected basically so we tend not to look at toer ended Evolution Dynamics I would say that active inference has uh some leverage to represent that that other the framework would not have because of the relationship between uh like constraints as a remal thing and beliefs as an intentional state yeah one thing that makes me think of so when we're when we're doing modeling based uh uh outcome based modeling whether for hobby or for grad school for a job like we're often in a setting that either expects or Rewards or accepts only a converging answer because uh because it's being you know does insulin in dropa change their circadian this or that it's like that's it's very it's very converged of a question already and it's something where we're looking to reduce our uncertainty about um so then that um closes off I mean to the extent that it converges it closes off Divergent routes not just of like doing and addressing that question that's more of a methodological point but it then we we don't um make models that can see a more open-ended Horizon of what could actually happen in the system because that always adds more State spaces that can be broken and unfolded so I guess again to the kind of Kernel of the question you've written these very interesting sequence of papers about symmetry breaking and and all of this so how do you use that in a way that goes beyond just stating larger and larger State spaces or feeling less and less confident when it seems like a large piece of this is that certain aspects of systems are unpre statable sorry confident about what precisely unpre statable unpressable yes I'm confident that uh you cannot preate the space of possibilities of space of belief that an agent can have and not devop any Evolution something that could like that is by definition of the terms something that could be argued is that there is no such thing as a plan Evolution but that would be a that would be a a very uh bold thing to claim and most people would not say that explicitly ever because it's kind of obviously not the case if there is an arent for it I'd like to hear it but the argument that um are uh stated against the upper andd Devolution thing are um um basically you have um I would say all rely on a confusion between the fact that the law of physics work which they do and the idea that the specific language that physics talks is a full and objective representation of everything there is or could be which is like it's a possibility we can discuss about it but this is a distinct possibility that the statement that the law of physics work which again they do okay cool well then just to sort of close on this active advantages many interesting conceptual advantages we hope analytical and more uh

transdisciplinary or more holistically and also another huge piece is it's connected to an always improving open-source computational environment we expect and prefer but in that case linking a system in year X social system or model it's kind of like using a Linux system that there could be some developments like in message passing algorithms or interoperability of different generative models or category theoretic augmentation of models that are connected to the active inference ontology all these different kinds of synthetic Technologies which we can basically already imagine and to some extent already have it becomes uh a way to map systems to um a lot and that that is not as far as I understand um plausible without kind of a first principles and the ecosystems of share intelligence and all these other kind of memes and themes that we talk about all the time surprise minimization things doing as most likely for them to do Paths of least action basium mechanics then there's a continuity of explanation with the rock just being there and the investor trading with this strategy an instrumental unifying account that makes relatively minimal claims about the territory that includes the Rock and the investing activity what' you add in uh but um like what you said message passing is uh interesting to me because you have a broad Coalition of physicist mathematicians or scientist people in Academia who basically think there is one language to Nature and what we do as scientist is to discover it and I I do not think that holds water not only because of an evolution is a thing uh which is a sufficient actually but if you look at Mage passing if you look at Quantum bism if you look at category Theory you open formally and conceptually the possibilities that there are different languages in the world and they may be more or less in their operation and they may be non-reducible to each other and then you start to ask Liv questions I'd say especially when uh implicitly or explicitly you are interested in what determines the fact that a specific language Works in a specific context why is um yeah I'm not not going to go further down the math road but you have important uh compatibility SL inter operability result between major fields of uh physics um I will give a reference by John B B I don't know you pronounce it I never this base so we have important interoperability results in uh fields of physics and formal science and yeah we don't have a proof however that um you know all possible presentation of physics are interoperative operable op how do you say that interoperable interoperable interoperable okay so from No it's it's it's a great point and what's really interesting is like the relationship between base graphs first off base graphs being analytically described and then computationally implementable is on the order of decades old and then as we kind of explored just yesterday in Liv stream 55.0 the relationship between the Bas graph and forny graphs which enabled message passing to be done procedurally to

optimize a base graph which just brings all of these computational all these analytical connections but that wasn't understood until it was and that was only slightly over two decades ago so then in fact those kinds of results and developments reflect the the changing social and scientific Niche so I don't know maybe this is a different question but science as a social activity and then I don't know does that how does that ground our work or give us more what so um about um the whole language thing uh I don't don't know where it's at now but at least 10 years ago it um seemed that you can have a message passing you can have well you can have Bas and coherence of state of observable that does not rely explicitly on the probabilities uh as in a disposition that is embedded in a specific set space you can have Bas in coherence that pre-exist to the representation in which probability is embedded and if there was a way to cross this and message passing You' have pretty straightforward straightforward um road to uh a straight account of contextuality and social construction I say this with SC quotes oh great great note exciting for um Quantum context and Quantum social and scripts and constraints and Norms let's close that question for now though nice okay um how can yeah great question thank you to whoever asked and again video description you can ask it'll appear and we we'll see if anyone writes more questions during the stream how can or does a framework Theory principle of everything articulate with moral normative ethical deontic considerations so I I think it is uh addressed in uh a good on of detail by an activists how you relate a theory of biological individuation to a theory of Ethics uh I do not have a good grip on this literature so you have um like how to say high level intuitive uh opinion on that but this is not like state-of-the-art and um okay basically I I uh I'd say this flows well with stoic virtue ethics things act according to the nature so you may as well work so that your nature is good and then you have a record I ity in counting what is good and what is not and I do not see um straightforward analytical way to resolve this ambiguity I said circularity or vity both apply yeah so I think Evan Thompson has worked n and may have a much better answer to how we relate theories of being and theories of physics oh verilla did actually yeah I guess my sort of short answer would there there would be F first just we would have to hear a lot more even possible perspectives but without even hearing them can at least maybe say that there is an articul and a meter stick as an instrument and this kind of goes actually to that question about the B spaces and uh what is preconditioned so in what ways is the meter stick um articulate with Downstream ethical Frameworks what ought to be done or just normative more broadly um there's Downstream decisions that the meter stick is agnostic to and then the meter stick is also a social result that is itself Downstream of other norms and constraints biotic abiotic

constraints regularities of space time I would say there is a um what we can do with a theory of sense making with active influence or in activism is to um like just describe what we mean by normativity Antics what are the underlying natural processes when I say natural what mean by opposition to social those processes are biological and social at the same time and that that may bring reflexivity or distance uh to account of normativity that may Pro prove some of them incoherent or uh ill advised that will not give you a solution to what is ethical and what is not notably because um like the most natural way to um derive ethics from something like an activism or something like inference is that you both those framework are centered around biological individuation so we may say that biological individuation is good actually that sound sensible but then the question is biological ination for whom because you have a right I guess or desire will at least to exist but SOS cancer cells within your body who trumps who I don't know not from a Pur account there is no like there is no way you can straightforwardly only derive a prescriptive notion of good from any physics any descriptive account of the world you will have when you do when you make normative claims you have goals because you are living thing and because of active inference SL activism we know that cognition is embedded intivity in the uh autonomy in the biological activity of the agent so the theory that you're framing is in some way motivated even if you don't know how or at need that it is and then I'd say from there it doesn't quite destroy the project of an objective systemics but it makes it much harder to defend so yeah I'd say you can derive a lot of negative results from descriptive theory of um like I wouldn't say ethics because ethics are non descriptive but descriptive theory of how ethics to be I would say that it is uh hard if not impossible to derive um like positive results if someone tried to do it I I'd like to see what this is wow thank you AO that last piece was very interesting many results many many ant colonies with their behavior to model many real situations to understand social constraints and Norms in instrumentally with descriptions that are then in the discussion section of paper and the data are the data and those are the descriptive aspects it's it's a different kind of data in a slight way but there's already enough to describe and interpret from what would be merely described and it is very hard to D derive positive results first it's a hard domain to make positive results in not not just ones that make people feel good but by positive like positivism like um Justified true assertions about a metaphysical space so it's hard to derive positive results about normativity and the kind of areas that this person asked these are all areas that are are um and it's hard to use a quantitative Theory which is almost it's an abstraction or it's just one language it's hard to use that to make positive results external to the system harder to make external to

the system positive results internal to the system are dime a dozen but then to really make that reflexive Loop that you described between the um the what is and what what we expect and prefer which is actually woven up in an interesting way in actm rather than two kind of diverge timelines like what we quote expect to happen like a prediction of the likeliest future and then a second timeline that's like well what what what I would really prefer to be doing in di the agent just has one expectation SL preference which loads up on the pragmatic value but that gives a unified approach towards one's expectations of how things will be and what they expect SL prefer them to be I would say that at the vibe check level active inference flows well with the idea that empowerment is good but make it a formation would be uh quite hard you could say that France Vibes well with the notion that everyone knowing their place is good which is quite the Opposites so I don't see an ambiguous relation between active inference and any um ethical framework I can think of I would say again that it uh like flows well with a virtue ethics framework because of the relation between uh one's action and one's nature according to AC inference nature being here a term that is like maybe not not the best term but like I'm I'm literally using a traduction of uh uh before Christ text so yeah nature is not that bad just not what we mean uh in the context of 19s 20s discussions yeah that's a very complex topic with with definitely a lot of Rich work on like our cities and people nature are humans animals and uh that sort of homo sapiens type discourse and then how does active inference frame the relationship with the niche in terms of entanglement or communication or co-action intraaction communication and in influence through blanket what where does that go beyond just describing it doesn't have to I think a sort of base level and what you referred to here is like that artifact the product of the work which is what the work is about can be a jumping off point for other directions and so the work of active inference modeling can be in what it leads to it's not always internal to the model that the improved awareness or better decision is realized uh I can see one quite straightforward and I don't think it is entirely broken way to relate ethics to active infen is uh A6 like not not A6 normativity as a social f it is a collective fomen it's about social relationships and you could apply uh descriptive theory of human behavior and like human social aggregated Collective Behavior Uh to find out what traits in US organization are prone to survival or adaptivity and you could say that those traits are uh good actually by construction of what we mean by ethics that could be a coherent claim and then you could be interested in what trades make a organization durable but I see this is the only logically right for why I see but that would like tend to um identify as good things that uh I would not say as good like theodism it lived quite a long time well theodism let us say um

old school Empires with no claim to representations or uh Power by the uh how do you say populace and you could uh integrate uh you could basically um renormalize this Theory not around uh clock time but around I don't know metric of Lifetime how much things happen within the organization and then you'd start having something that is less obviously wrong and you could have a basically eics be about optimization of specific parameters that can be addressed by descriptive framework but you'd have to make a lot of jumps to get there and while I do not see an obvious problem to that it is very possible that we would not like the conclusion of such optimization thing and again if you want to model something you have to decide what are the parameters of interest and and you can be wrong about that so here again you do not have an ambiguous even at a purely descriptive level you do not have an ambiguous application of anything to anything would say that more empowerment is good actually and I would say this goes that many nested scales of organization and I would say it as somehow something to do to the physics of biological individuation but only in a Vibe check way nothing wrong with that um yeah especially with some of the recent work like the variational synthesis with Eevo and the connections between the statistical concepts of persistence or thinkness from free energy principle related to to remeasurement of the Observer or the organism of them that concept of repeated measurement being compatible in our work and and in other related Works basically with Fitness if there's a height distribution of trees in the forest and then a certain time Point later the height distribution is the same that was stabilizing Dynamics we could say stabilizing Evolution or stabilizing selection if it moved it moved and so how do since there is such a coupling between things being as they are and basically what the distribution of what is there is described both this stoic way but also this kind of evolutionary biology way so does of inference offer any other or complimentary perspective other than just the kind of strengths and weaknesses of the of using Fitness phenotypic Fitness or mtic Fitness as its own justification to say that is to say that something is good because it exists this is not a strong position to take so this is why you'd have quite high level of difficulty to um uh derive anything prescriptive from anything descriptive the function is not the same you can make relation between the two no eics is purely um blind about nature no um the of nature is purely ethically unmotivated but to have formation between the two is probably not possible at least highly non-trivial okay so to kind of close on question how does a free energy Principle as a principle for every and active inference as an implementable framework for applying free energy principle systems how does that articulate with these broader very um deep and fundamental Human Social context short answer seems to be it's a

lot like other scientific ideas inherits a lot of their challenges and frictions and and situations also there appeared to be some interesting promising routes or themes to weave in however even with a very charitable or optimistic interpretation there seems some in principle challenges with these areas that make certain kinds of claims perhaps not even within the proximate understanding we have of active inference but people should totally write in the live chat or add comments and stuff I mean it's just interesting question okay how are higher order beliefs about doing better than expected at PM probably prediction error minimization integrated into the architecture of the agent's internal model of external Dynamics and itself in those Dynamics my thought here is that we have prior beliefs to fulfill certain self-evidencing expectations EG homeostatic predictions but how do those interact with these higher order predictions which appear to my mind to deny or negate the fundamental priors in so far as they predict the agent to go over and above the levels of prediction error minimization which otherwise might be achieved by fulfilling those fundamental priors they gave their thought let's return to the original question I do not know like I know that um a comparison to a baseline homeostatic level of tationation is a quite important thing in a patry notably in the work of bars I do not know how that's implemented actually in computational models so I'd rather not answer approximatively uh Ben White should be more informed about these things yeah or maybe you well I mean the word that seems to be doing the work is integrated so higher beliefs about doing better than expected at prediction error minimization or we can just say free energy minimization so this was explored in the affective inference work so if the um free energy was um decreasing as expected it was associated with neutral veilance if things were improve proving better than expected it was associated with positive veilance and if things were doing worse than expected it was negative veilance so because of the ability to make hierarchical otim models and just hierarchical basian models in general it is possible to talk about these beliefs about beliefs metacognitive beliefs so it is possible to make models that have expectations about how well even an optimization process is going to go and then a computational psychiatrist or whatever May associate that with the affect how is that integrated into the agent's internal model of external Dynamics well shortest answer is to sketch it out how one sees it and then it becomes either there or not but just in general if the internal Model includes these higher order beliefs about doing better than expected then that's the agents generative model that then interfaces or articulates with external Dynamics so you could have some environment that has a environmental regularity or Dynamics or that are influenceable or or not and then you could deploy two agents one of them that has the

Baseline generative model and the other one has this doing better than expected layer that the questioner is asking about and then they can do parameter sweeps and maybe adding this higher order belief increases the state space a lot and so it increases the computational cost a lot but some interesting pockets of behavior are found or the behavior in that setting ends up looking the exact same but in another setting it would have been different a lot of these questions Point towards very important questions which can basically only be pointed at because so much is going to depend on what the actual model is okay um where can I find the recommended reading list mentioned in the lectures so where we have the um slides from the lecture they're in live streams just go to the big table and find the slides if you find a lecture that doesn't have slides just let us know and we'll we'll add it um where things were just mentioned in the lectures that's a great question and our goal at the Active inference Journal Beyond just transcribing and Publishing and adding captions and translating and everything of these live streams we would like to also do more bibliographic work so that the books that people mention we can also have in that open-source versioned Journal so we generate the first version of transcript people update it improve it oh what was the name of that author what book are they talking about here and then we can actually have a unified bibliography section for the Stream So if that sounds interesting to you and you'd benefit from it then surely many many others will too so if you're excited to do that kind of active inference Journal work I highly encourage SL recommend it I what else will you at same I kind have nothing to yeah well we can look at some of the other questions that were added that others uh wrote people in live chat can add question or they can go to the um course website and ask a new question do you want to look at one of the previous ones or do you have any other Just Thoughts you want to bring up not really the first ones were on sword so I don't not like and they are they they were asked weeks ago so there is a relatively low likelihood that going over them again would be very helpful I think we can cut the discussion cool but the important things that are not treated now can be Tre in conclusion anyway um okay so what are the next steps then for the course next step is did a lot of people register for the uh presenting in the session I haven't gotten any specific responses or like titles yet okay so we see if people uh do that and yeah we can write write another message um including even a short video message or just a written one and then also ask specific people also if anyone is like interested to present something but they're not really sure the format or they want to pre-recorded we're just kind of exploring and seeing what the people who are following along live want to do and also trying out things that

we think Futures will want to do too so if someone has an idea on that we're definitely open to any last thoughts nope I'm good cool all right thank you then see you all bye